

Problem Solving in Geometry (III)

07.10.2023

1. **Warm Up** $ABCD$ is a square. E, F, G, H are points on AB, BC, CD and DA respectively. The lines HE and GF intersect at a point P . If HF is perpendicular to EG , show that

$$PE^2 + PH^2 = PF^2 + PG^2.$$

2. Let $ABCD$ be a parallelogram and let P be a point inside it such that $\angle PDA = \angle PBA$. Let ω_1 be the excircle of PAB opposite to the vertex A . Let ω_2 be the incircle of the triangle PCD . Prove that one of the common tangents of ω_1 and ω_2 is parallel to AD .

3. Consider two concentric circles Ω and ω . The chord AD of the circle Ω is tangent to ω . P is a point on the minor arc of AD of Ω . The tangent lines drawn from the point P to the circle ω intersect Ω again at points B and C . The segments BD and AC intersect at the point Q . Prove that the segment PQ divides the segment AD into two equal parts.

4. Let I be the incentre of acute-angled triangle ABC . Let the incircle meet BC , CA , and AB at D , E , and F , respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.